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1. Abstract
2. Introduction
3. Method
4. Results
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Introduction

Researchers and scholars alike seek funding support to advance innovative and creative approaches to real-world topics. Funding, to name a few broad areas, helps pay for materials, time, consultants, as well as reducing dissemination costs either for presentation or manuscript submission. One common method for funding support is through the submission of grant applications. A challenge for any funder once these applications are received is how those applications can be reviewed in a **timely, fair, and equitable** manner [Guthrie et al., 2019]. Yet besides just those two tenets, it also requires the identification of specialists to review, program officers to train these specialists, and then a method of scoring and feedback that is transparent and provides the applicant with sufficient information as to why their proposal was funded, funded with revisions, or not funded [Van Rooyen et al., 1999; NIH Report, 2024]. Studies have consistently identified issues like **bias** (e.g., against gender, age, or institutional prestige) [Martin et al., 1986; Hultgren et al., 2025] and the burden/inconsistency of human review as persistent limitations of the peer review system.

A recent advance and societal tool that is being implemented in many situations is the use of **Artificial Intelligence (AI)**[Goldfarb & Teodorescu, 2023]. Large language models (LLMs) have the ability to analyze data and provide responses based on prompts that are given [Brown et al., 2020]. LLMs continue to improve based on the amount of data that is provided. The advantages of using AI and LLMs are often cited as their potential to provide **objective, data-driven recommendations** [IBM Policy Lab, 2023] and operate **relatively quickly** when processing large volumes of text [Shrotri & Gupta, 2024]. This efficiency is demonstrated in tasks like automated screening and filtering of scientific literature [Gao et al., 2024], offering a mechanism to counteract human subjectivity and review burden.

Complementing human specialist reviews with AI grant reviews to help assess grant submissions has the potential of providing decision-makers with more data points to render funding decisions and the potential for **robust feedback** to applicants [LLM in Peer Review Study, 2025]. Implementing this adjunctive technological reviewer requires an understanding about how much in agreement or discordance an AI grant reviewer would assess an application. This exploratory pilot study delved into this topic by implementing three experiments. Using an existing grant application opportunity we assessed proposals comparing human reviews to native AI reviews (Experiment 1); comparing human reviews to AI reviews that were trained with expectations and samples (Experiment 2); and repeating Experiment 2 with the explicit prompt to AI to score stricter (Experiment 3).

Updated Reference List

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Methods

Study Design

We conducted a three-condition, multi-vendor repeated-measures comparative evaluation in which three commercial LLMs reviewed faculty seed-grant applications under different prompt

engineering strategies and were compared against human expert reviewers using the same scoring rubric.

Setting and Context

The School of Health Professions at Stony Brook University provides opportunity for faculty to apply for seed grant funding as a way to support scholarship agendas. In 2025 an open call was distributed to faculty members in the School of Health Professions. Faculty who were interested were provided with instructions to follow the template and to submit their proposals. Once proposals were submitted they were then distributed to 4 reviewers.

Reviewers were provided with a rubric for scoring and providing subjective comments for each application (Figure 1). Scoring was completed in 6 areas: Innovation and Impact (30 points), Research Methods (30 points), Team Qualifications (10 points), Funding Potential (10 points), Budget (10 points) and Writing Quality (10 points). In addition a final question was asked with a summative disposition from the reviewer of Fund, Fund with Revisions, or Do Not Fund. Reviewer data was collated and used to present to the Dean of the School for final funding decisions.

Following the funding decisions, the submitted proposals were then processed through three different LLM's. For this study we selected three LLMs: Open AI's ChatGPT, Google's Gemini and X's Grok were utilized (Figure X). These three LLMs were selected because of their popularity and ease of access, in addition to their application programmable interface (API) to allow for faster experimentation. Three different experiments were then employed utilizing each of these LLMs and compared to human reviewers.

Three Experimental Conditions

This study tested three distinct experiments, each employing a different prompt engineering strategy corresponding to common approaches in LLM application: zero-shot learning (no training examples), few-shot learning with distributional information (multiple training examples showing score variance), and few-shot learning with stricter prompt instructions (multiple training examples combined with more explicit conservative guidance).

Experiment	Experiment Name	Applicant	Human Count	Human Reviewers	LLM Count	LLM Breakdown
Exp1_ZeroShot	baseline_v1	CHRISTINA	4	AM, HW, SW, Y	9	Google=3, OpenAI=3, xAI=3
Exp1_ZeroShot	baseline_v1	KELLY	4	AM, HW, SW, Y	9	Google=3, OpenAI=3, xAI=3
Exp1_ZeroShot	baseline_v1	DANIEL	4	AM, HW, SW, Y	9	Google=3, OpenAI=3, xAI=3
Exp3_FewShot	multi_examples_v1	CHRISTINA	4	AM, HW, SW, Y	9	Google=3, OpenAI=3, xAI=3

Exp3_FewShot	multi_examples_v1	KELLY	4	AM, HW, SW, Y	9	Google=3, OpenAI=3, xAI=3
Exp3_FewShot	multi_examples_v1	DANIEL	0	None	0	Google=3, OpenAI=3, xAI=3
Exp4_Stricter	strict_scoring_v1	CHRISTINA	4	AM, HW, SW, Y	9	Google=3, OpenAI=3, xAI=3
Exp4_Stricter	strict_scoring_v1	KELLY	4	AM, HW, SW, Y	9	Google=3, OpenAI=3, xAI=3
Exp4_Stricter	strict_scoring_v1	DANIEL	0	AM, SW, Y	0	Google=3, OpenAI=3, xAI=3

A critical methodological consideration across all experiments was preventing data contamination when using human reviews as training examples. For experiments requiring training examples (Experiments 3 and 4 described below), we purposely selected DN's application and the four associated human reviews as the source of training data through random assignment from the three available applications to minimize selection bias. DN's reviews were particularly suitable for training purposes because they score variance across the four human reviewers, with scores ranging from more critical to highly favorable assessments. This distributional characteristic made these reviews ideal for calibrating LLMs to the full range of the scoring scale as they included expert disagreement.

Experiment 1: Zero Shot Baseline

Experiment 1 established baseline LLM performance without calibration (e.g., training) examples, representing the most common deployment scenario where LLMs receive only task instructions without training data. The prompt included standard instructions describing evaluation criteria, a scoring rubric with point allocations, and a request for criterion-level scores and overall recommendation. No training examples were provided.

This experiment generated 27 LLM reviews (3 vendors × 3 applicants × 3 iterations each) for comparison against all 12 human reviews from the study (4 reviewers per applicant across the three applications: CR, KL, and DN). Since no human reviews were used as training examples in this zero-shot condition, all human reviews remained independent of the LLM inputs and could be included in the comparison group.

This experimental design allowed us to assess whether LLMs could perform grant review tasks using only the evaluation rubric and application content, without the benefit of example reviews to calibrate their scoring.

Experiment 3: Few-Shot Learning

Experiment 3 tested whether providing multiple training examples with natural score variance would improve LLM output across the full range of the scoring scale. This experiment was

motivated by the hypothesis that LLMs might exhibit optimism bias when reviewing grant applications, and that exposing them to examples of legitimate expert disagreement could help tune their scoring to match the realistic distribution of human reviewer perspectives.

The prompt structure built upon Experiment 1's base instructions, adding all four complete human reviews of DNs application as training examples. These four reviews represented the natural distribution of reviewer perspectives, including high-scoring, medium-scoring, and more critical assessments of the same proposal.

This condition generated 18 LLM reviews (3 vendors $\times$ 2 applicants $\times$ 3 iterations each) limited to CR and KLS applications only. The human comparison group consisted of 8 human reviews, specifically the 4 reviews each from CHRISTINA and KELLY's applications. All four human reviews of DNs application were excluded from the comparison group because they had been provided to the LLMs as training examples. Additionally, all LLM reviews of DNs application were excluded to prevent unfair comparisons where LLM performance on an application with training data (DN) would be compared against applications without training data (CR and KLS). This exclusion strategy ensured that the comparison group consisted only of reviews for applications that remained independent of the training data for both humans and LLMs.

Experiment 4: Few-shot Learning with Strict Prompt Instructions

Experiment 4 tested whether explicit conservative instructions could counter potential optimism bias in LLM grant reviews when combined with multiple training examples. This experiment investigated whether adding explicit directive language emphasizing critical evaluation to the few-shot approach would reduce any tendency toward lenient scoring. The prompt structure used the same few-shot learning approach as Experiment 3, providing all four complete human reviews of DNs application as training examples, but supplemented these with explicit instructions to apply strict and conservative scoring standards. The additional calibration instructions included multiple directives designed to encourage critical evaluation:

"The above examples show that expert reviewers can disagree substantially (61-80 point range) on the same proposal. However, they share a common tendency toward critical evaluation. Your task is to match this level of critical evaluation. Be conservative, identify weaknesses clearly, and do not inflate scores. Use these examples to understand the full range of the scoring scale and the appropriate level of scrutiny."

Like Experiment 3, this experiment generated 18 LLM reviews (3 vendors $\times$ 2 applicants $\times$ 3 iterations each) limited to CR and KLS applications only. The human comparison group consisted of 8 human reviews, comprising the 4 reviews each from CR and KLS applications. All four human reviews of DNs application were excluded from the comparison group because they had been provided to the LLMs as training examples, maintaining the same exclusion logic as Experiment 3. Similarly, all LLM reviews of DANIEL's application were excluded to prevent comparing LLM performance on DN (the source application for training data) against applications without training data (CR and KLS).

Results

Overview

We generated 63 LLM reviews across three prompt engineering experiments (27 reviews in Experiment 1, and 18 reviews each in Experiments 3 and 4) and compared them to human expert reviews using experiment-specific comparison groups to prevent training data contamination. Results are organized by experiment, with each section presenting overall performance, criterion-level analyses, vendor comparisons, recommendation agreement, and consistency metrics. Cross-experiment comparisons follow the individual experiment analyses.

By Experiment

Table X below summarizes the the descriptive statistics for each experiment, including the number of human and LLM reviews used in comparisons:

Experiment	Exp1_ZeroShot	Exp3_FewShot	Exp4_Stricter
Human n	12	8	8
LLM n	27	18	18
Human Mean	79.0833	83.125	83.125
Human SD	13.3243	13.7263	13.7263
LLM Mean	83.6296	80.5556	72.4444
LLM SD	5.74555	5.36023	5.33823
Difference	4.5463	-2.56944	-10.6806
p-value	0.141244	0.492609	0.00786515
Cohen's d	0.443094	-0.246593	-1.02558
Broad Agreement Rate	33.3333	50	50
Exact Agreement Rate	0	0	0
Google Mean	88.4444	82.3333	78
OpenAI Mean	77.3333	76	69.8333
xAI Mean	85.1111	83.3333	69.50

Experiment 1: Zero-shot Baseline

In the zero-shot baseline condition, we compared 27 LLM reviews (3 vendors × 3 applicants × 3 iterations) to all 12 human reviews, as no training examples were used. Human reviewers scored applications with a mean of 79.08 ± 13.32 points out of 100, while LLMs produced higher scores with a mean of 83.63 ± 5.75 points. This difference of +4.55 points represents

approximately 5% optimism bias, though it did not reach statistical significance ($t(37) = -1.503$, $p = 0.141$, Cohen's $d = 0.443$). The effect size was small-to-medium, suggesting a tendency toward more generous scoring in the absence of calibration examples. Figure 1 reveals that LLMs not only scored higher on average but also demonstrated substantially lower variability ($SD = 5.75$) compared to humans ($SD = 13.32$), with fewer outliers.

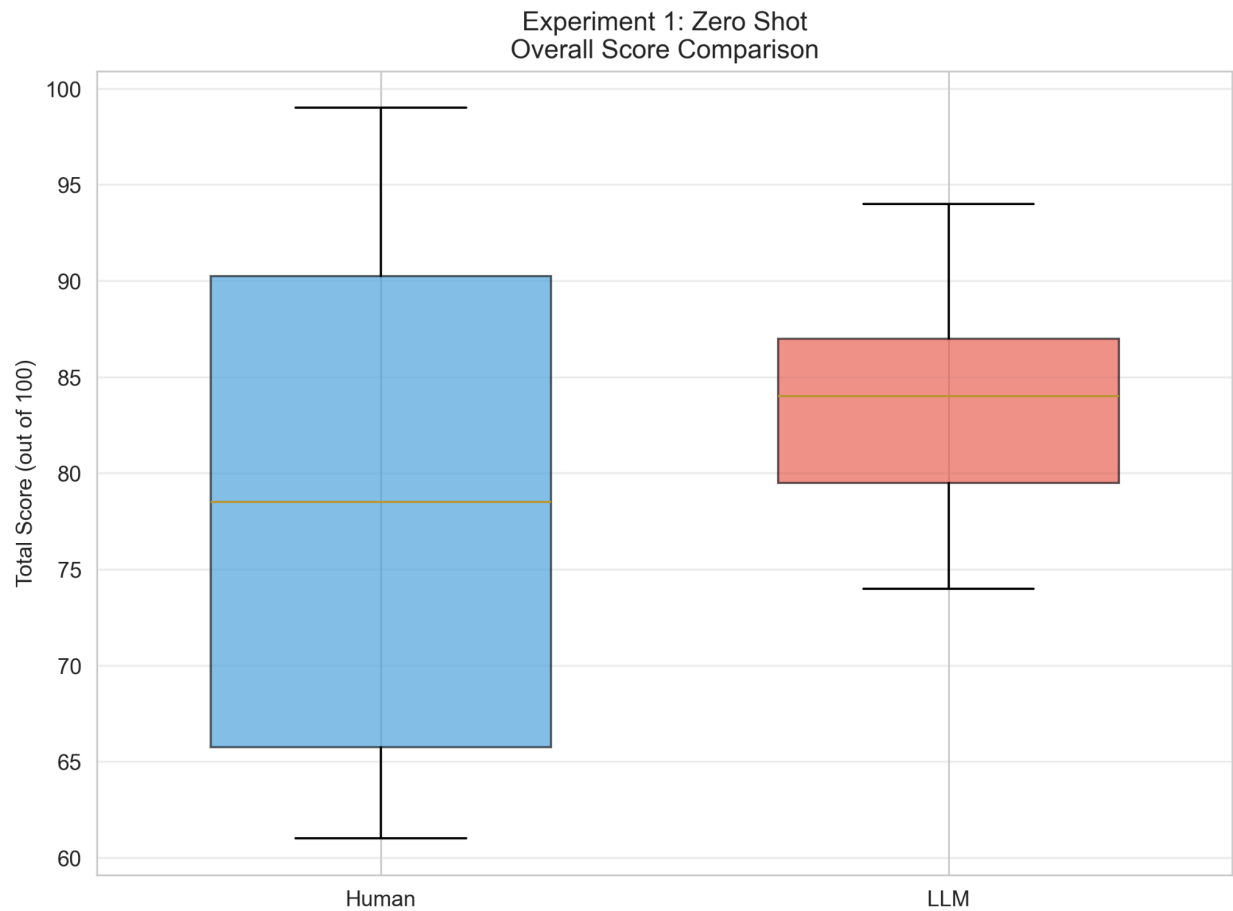
Criterion Level Analysis: The LLMs demonstrated significantly elevated over-scoring on External Funding Potential, rating applications 1.11 points higher than humans ($p = 0.004$), and Methodology & Feasibility (+2.81 points, $p = 0.073$). These findings suggest that LLMs may be particularly optimistic about practical and methodological aspects of proposals while showing more balanced judgment on conceptual elements like innovation. While Innovation & Impact showed only modest over-scoring by LLMs (+0.69 points, $p = 0.367$), as did Team Strength (-0.10 points, $p = 0.773$), Budget Clarity (+0.04 points, $p = 0.950$), and Presentation Quality (-0.01 points, $p = 0.981$). Figure 2 shows performance across all six criteria expressed as percentage of maximum possible score, and shows the elevation in External Funding Potential and Methodology scores for LLMs compared to human reviewers.

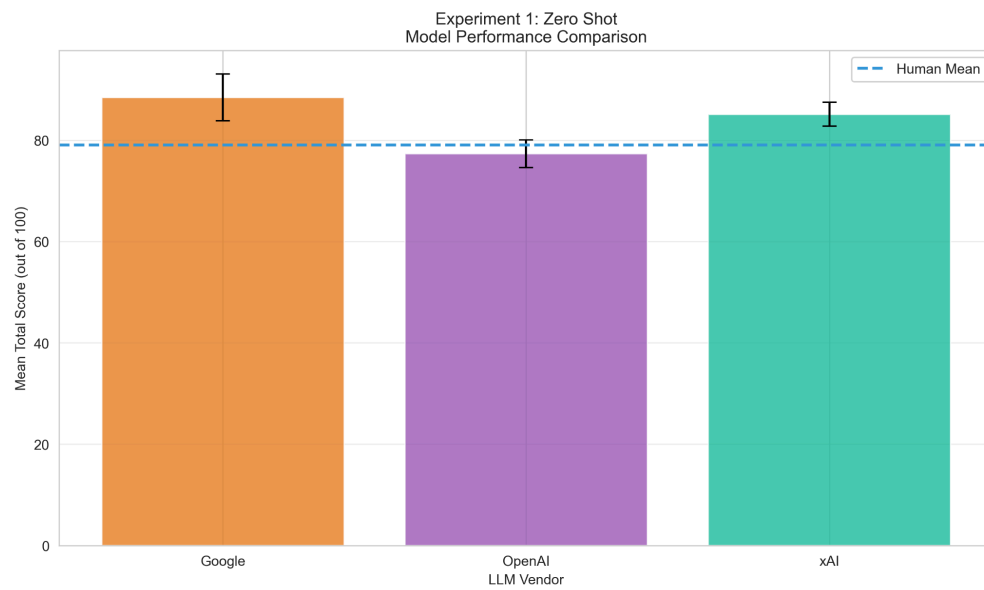
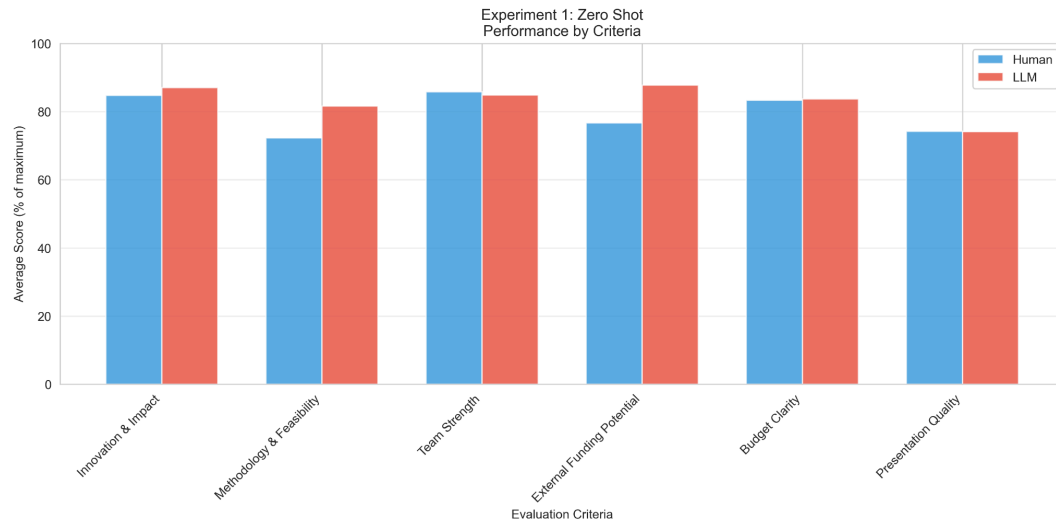
Vendor LLM Performance: There was considerable heterogeneity in scoring behavior across LLM vendors. OpenAI GPT-4 produced the most conservative scores (mean = 77.33, -1.75 from human mean), actually scoring slightly below human reviewers on average. xAI Grok showed moderate optimism (mean = 85.11, +6.03 from human mean), while Google Gemini demonstrated the strongest optimism bias (mean = 88.44, +9.36 from human mean). A one-way ANOVA did not detect statistically significant vendor effects ($F(2,24) = 0.87$, $p = 0.431$), likely due to within-vendor variability and small sample sizes per vendor. The horizontal line in Figure 3 represents the human mean score (79.08 points), against each vendor's alignment with human judgment.

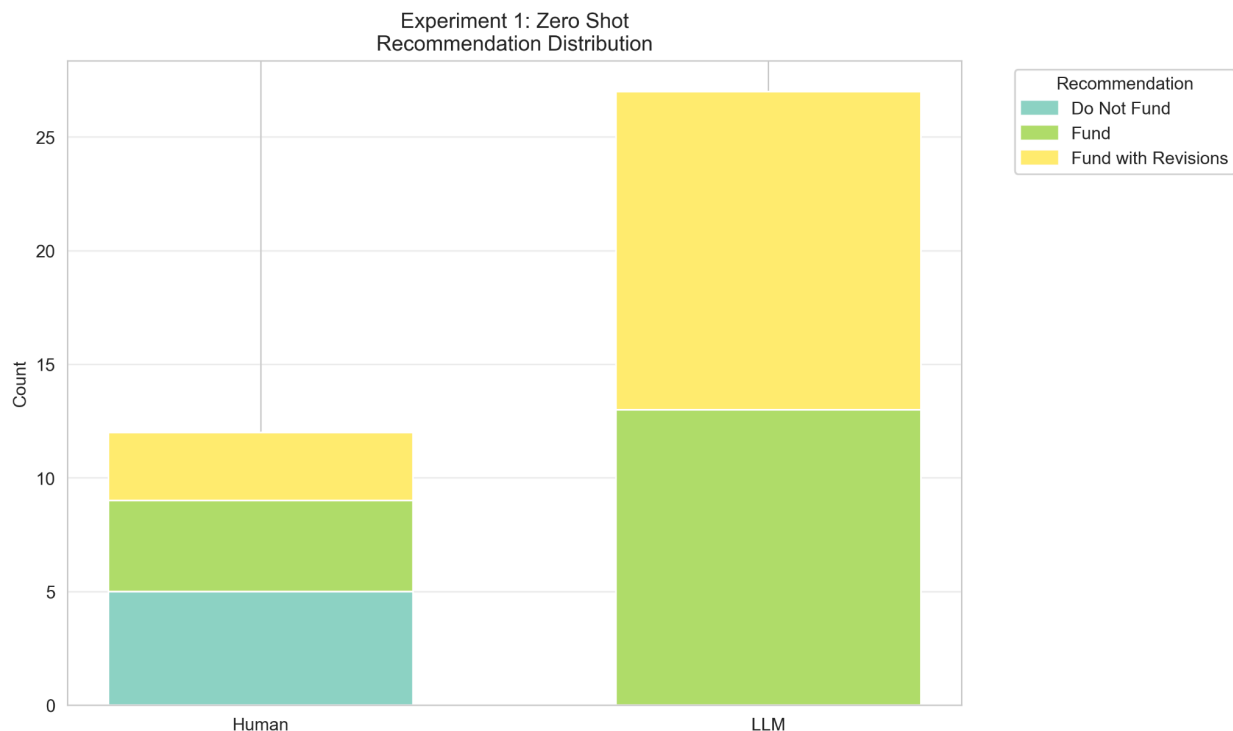
Recommendation Agreement: Human reviewers distributed their recommendations relatively evenly: Fund (33.3%), Do Not Fund (41.7%), and Fund with Revisions (25.0%). In stark contrast, LLMs showed a strong preference for intermediate recommendations: Fund (48.1%), Do Not Fund (0.0%), and Fund with Revisions (51.9%). Notably, LLMs generated zero "Do Not Fund" recommendations in the zero-shot condition, suggesting reluctance to make decisive negative judgments. We also evaluated agreement at two levels: broad agreement (grouping "Fund" and "Fund with Revisions" as positive vs. "Do Not Fund" as negative) and exact agreement (requiring identical recommendations). The broad agreement rate was 33.3%, while exact agreement rate was 0.0%, indicating complete misalignment at the decision level despite reasonable score alignment. Figure X illustrates the LLM tendency to avoid extreme recommendations in favor of the middle "Fund with Revisions" category.

Inter-Rate Consistency: LLMs demonstrated substantially higher inter-rater consistency than human reviewers. The average standard deviation across the three applications was 5.78 points for LLMs compared to 11.37 points for humans, representing a 49% reduction in score dispersion. This higher consistency could reflect either superior reliability or reduced sensitivity to genuine differences between applications. Correlation analyses between human and LLM

average scores by applicant showed moderate non-significant associations (Pearson $r = 0.664$, $p = 0.538$; Spearman $\rho = 0.500$, $p = 0.667$), limited by the small sample size ($n=3$ applications).







Experiment 3: Few-shot Learning

Experiment 3 tested the hypothesis that exposure to multiple training examples would produce stronger alignment between LLM with human reviews. We provided LLMs with all four human reviews of DANIEL's application for training, which represented scores ranging from 68 to 92 points, thus representing the varying disagreement between our human subject matter expert reviewers. For this analysis, we compared the 18 LLM reviews outputted to the remaining 8 human reviews of CHRISTINA and KELLY only, as all DANIEL reviews served as training data. Human reviewers scored applications with a mean of 83.13 ± 13.73 points, while LLMs produced a mean of 80.56 ± 5.36 points. This represents a non-significant difference between human and LLM reviews ($(t(24) = 0.697, p = 0.493, \text{Cohen's } d = -0.247)$), and an absolute difference of 2.57 points (~3%), the best alignment achieved across all three experiments. Figure X presents the score distributions for Experiment 3, showing the near-overlap in median values between LLMs and human reviews, a clearly improved calibration in scores versus Experiment 1.

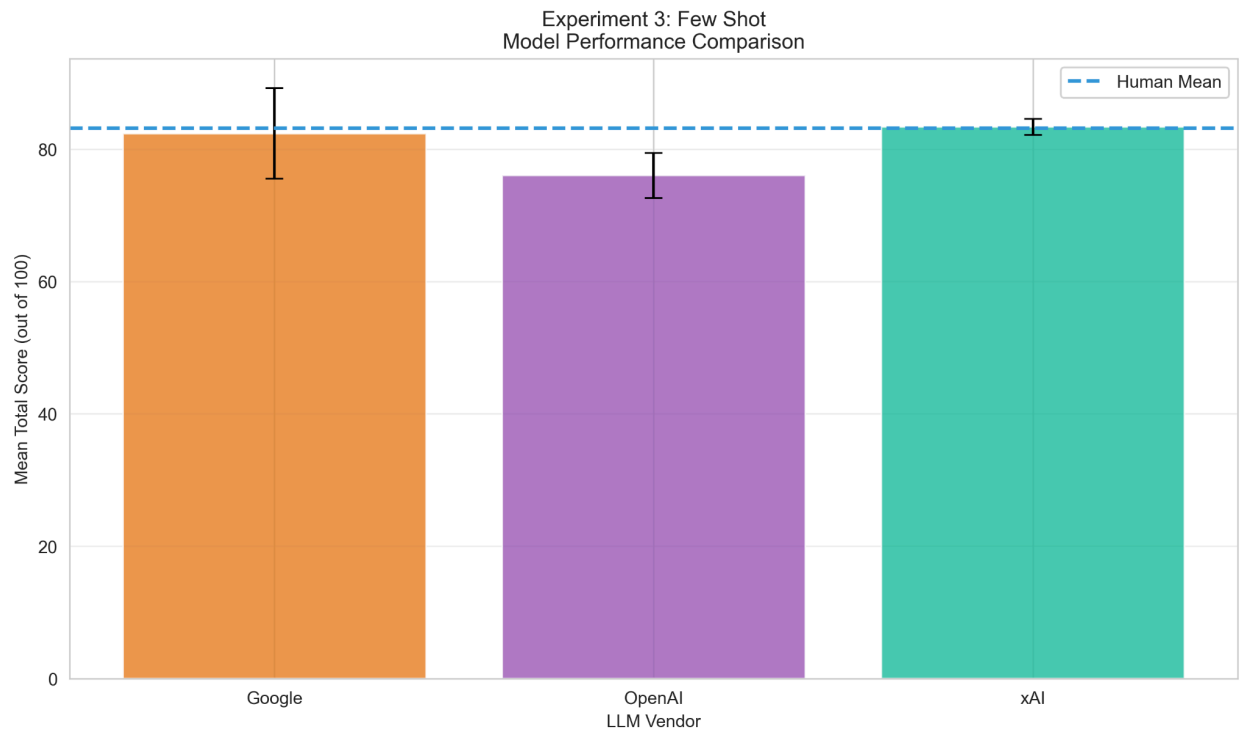
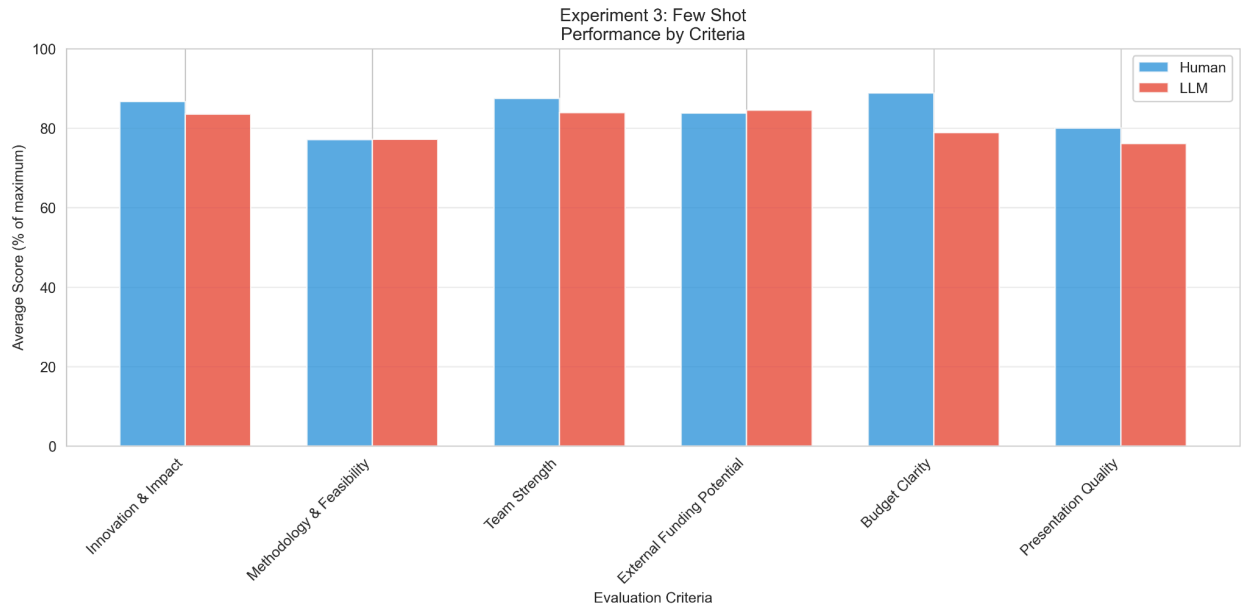
Criterion Level Analysis: Analysis of individual evaluation criteria revealed balanced performance, representing a major improvement over Experiment 1. The “Innovation & Impact” category showed modest LLM under-scoring (-0.94 points, $p = 0.260$), reversing the over-scoring observed in Experiment 1. While “Methodology & Feasibility” demonstrated slight over-scoring (+0.04 points, $p = 0.983$) in comparison to the consistent over-scoring in Experiment 1 (+2.81), and the “External Funding Potential” displayed near-perfect alignment

(+0.07 points, $p = 0.853$) compared to the significant over-scoring in Experiment 1. Team Strength (-0.36 points, $p = 0.516$), Budget Clarity (-0.99 points, $p = 0.144$), and Presentation Quality (-0.39 points, $p = 0.451$) all showed modest under-scoring, with none reaching statistical significance. Overall, no individual criterion showed statistically significant differences, representing balanced criterion-level output across all six dimensions. Figure X illustrates the normalized criterion-level performance for Experiment 3, showing near perfect parallelism with no extreme deviations.

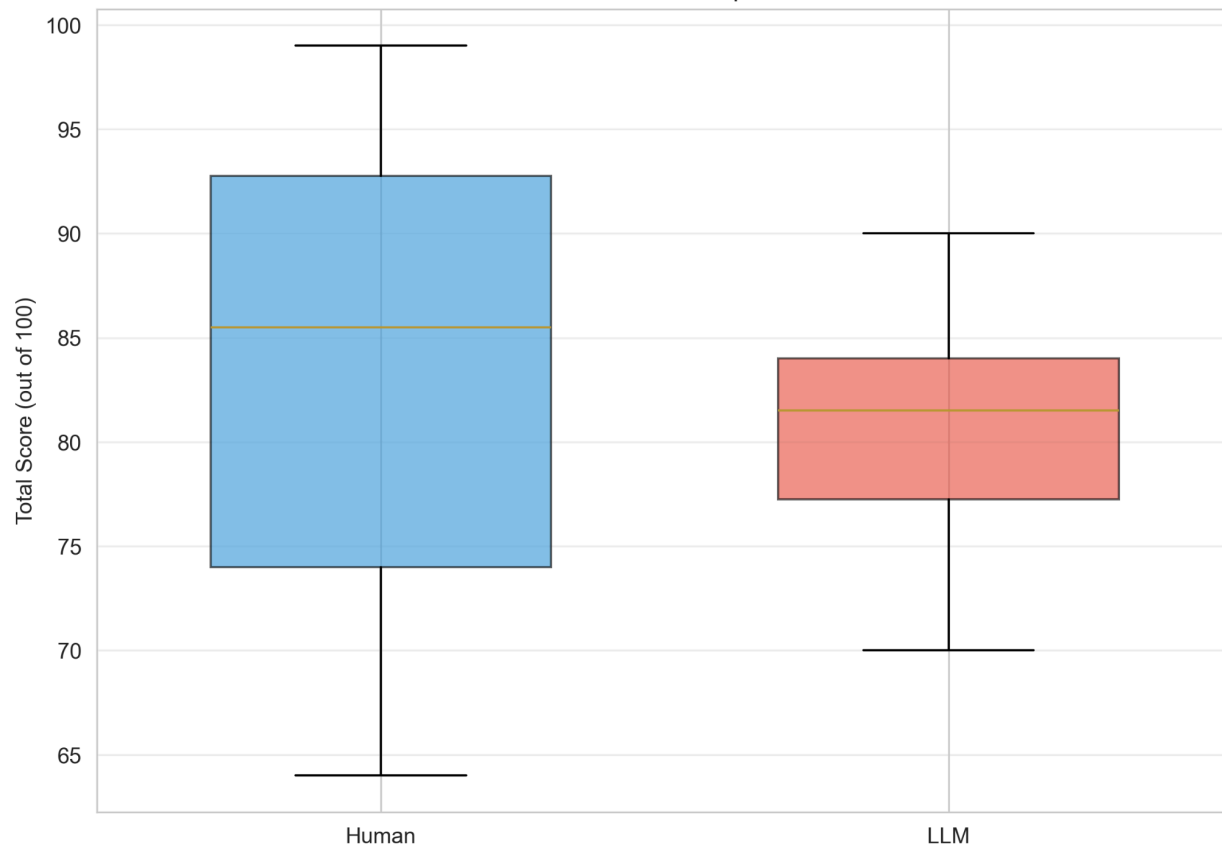
Vendor LLM Performance: All three vendors shifted toward more conservative scoring in the few-shot condition. xAI Grok produced scores closest to human mean (mean = 83.33, +0.21 from human mean), representing the best vendor-experiment combination observed in this entire study. Google Gemini, which showed strong optimism bias in Experiment 1 (+9.36 from human mean), now scored more conservatively (-0.79 from human mean), demonstrating that multiple training examples effectively recalibrated its scoring behavior. OpenAI GPT-4 showed the most conservative response to the few-shot examples (-7.12 from human mean), over-correcting from its near-perfect Experiment 1 alignment. These patterns suggest that few-shot learning impacts all vendors, but differently, in that they respond with different sensitivities to this newly provided information. Figure X displays the vendor performance in Experiment 3, showing all three vendors cluster near the human mean.

Recommendation Agreement: Despite the improvement in score alignment, recommendation agreement remained problematic. Human reviewers in this comparison group (CHRISTINA and KELLY only) showed a spread of all recommendation categories: Fund (50.0%), Do Not Fund (25.0%), and Fund with Revisions (25.0%). However, the LLM outputs showed even more extreme preference for the middle category compared to Experiment 1 with Fund (16.7%), Do Not Fund (5.6%), and Fund with Revisions (77.8%). Just as in Experiment 1, we also evaluated agreement at two levels: broad agreement (grouping "Fund" and "Fund with Revisions" as positive vs. "Do Not Fund" as negative) and exact agreement (requiring identical recommendations). The broad agreement rate was 50.0%, while the exact agreement rate was 0.0%. This dissociation between excellent score alignment and moderate broad agreement suggests fundamentally different calibration challenges. Figure X shows the stronger preference for "Fund with Revisions" in Experiment 3 compared to Experiment 1.

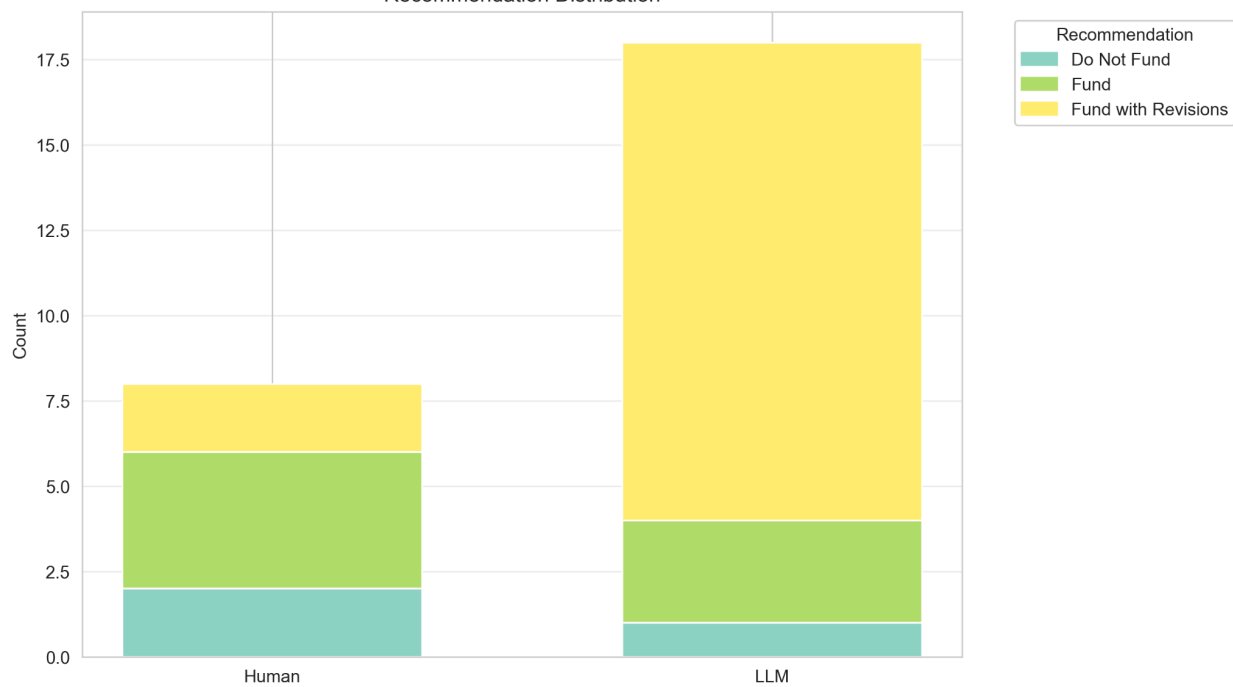
Inter-Rate Consistency: LLMs demonstrated higher inter-rater consistency in Experiment 3 compared to the zero-shot baseline Experiment 1. The average standard deviation across the two evaluated applications (CHRISTINA and KELLY) was 5.18 points for LLMs compared to 12.56 points for humans, representing a 59% reduction in score dispersion (this is the highest consistency observed across all experiments). This suggests that exposure to multiple training examples showing score variance may increase LLM consistency, potentially by clarifying evaluation standards that were not clear within the human provided prompt. Correlation analyses between human and LLM average scores by applicant were limited by the small sample size ($n=2$ applications after excluding DANIEL), precluding reliable statistical inference.



Experiment 3: Few Shot
Overall Score Comparison



Experiment 3: Few Shot
Recommendation Distribution



Experiment 4: Few-shot Learning with Strict Instructions

Experiment 4 tested whether explicit instructions to apply strict, conservative standards could counter the optimism bias observed in Experiments 1 and 3, when combined with few-shot learning. Using the same training examples as Experiment 3 (all four human reviews of DANIEL's application), we added explicit instructions such as "Apply high standards; this is a competitive program," "Reserve high scores for truly exceptional proposals," and "Be critical in your assessment; identify weaknesses clearly."

We compared 18 LLM reviews to 8 human reviews (excluding all DANIEL reviews used as training). Human reviewers scored applications with a mean of 83.13 ± 13.73 points, while LLMs produced substantially lower scores with a mean of 72.44 ± 5.34 points. This difference of -10.68 points represents statistically significant under-scoring ($t(24) = 2.900$, $p = 0.008$, Cohen's $d = -1.026$), with a large effect size. The strict instructions caused a dramatic over-correction of 15 points, swinging from the baseline optimism bias of +4.55 points to significant pessimism of -10.68 points. Figure X presents the score distributions for Experiment 4, showing LLM scores are substantially and significantly lower than human scores.

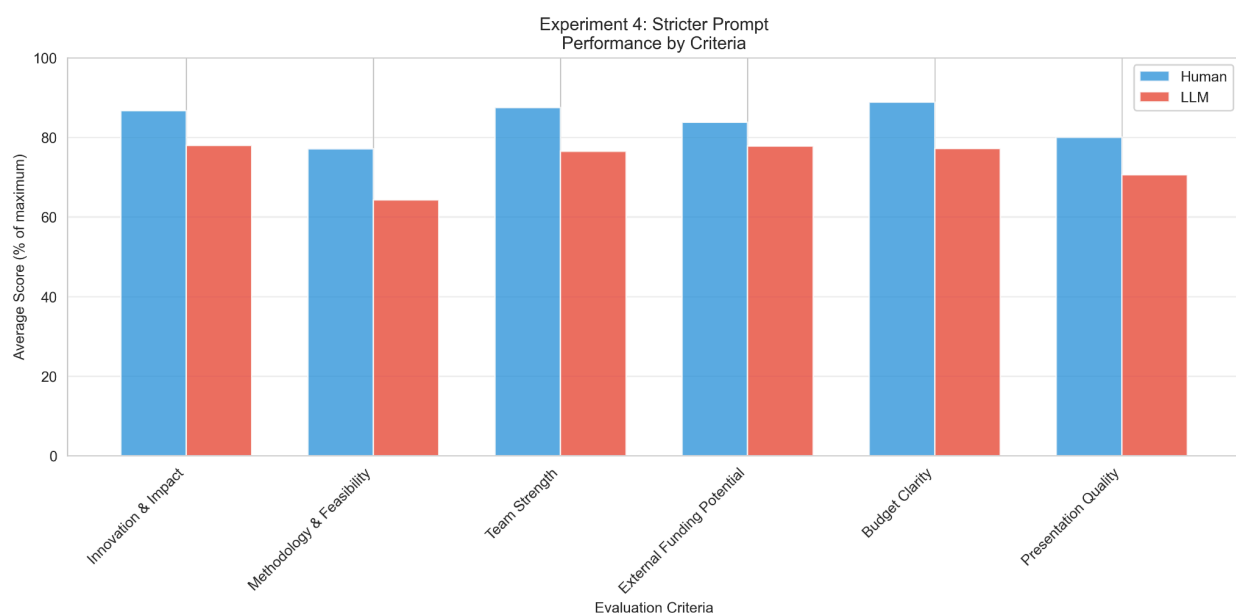
Criterion Level Analysis: "Innovation & Impact" showed statistically significant under-scoring (-2.61 points, $p = 0.003$), suggesting that instructions to "be critical" led models to undervalue novel contributions. The "Presentation Quality" also suffered, with LLMs scoring applications 0.94 points lower than humans ($p = 0.047$), representing an under-valuation of overall grant clarity. While "Methodology & Feasibility" (-3.85 points, $p = 0.066$) and "Budget Clarity" (-1.15 points, $p = 0.090$), and Team Strength (-1.10 points, $p = 0.074$) showed modest, non-significant under-scoring. Finally, "External Funding Potential" which had been systematically over-scored in Experiment 1 now showed only slight under-scoring (-0.60 points, $p = 0.198$). Overall, this pattern suggests that LLMs interpreted "be strict" most literally for subjective, qualitative criteria while maintaining relative consistency on objective, quantitative dimensions. Figure X displays the normalized criterion-level performance for Experiment 4.

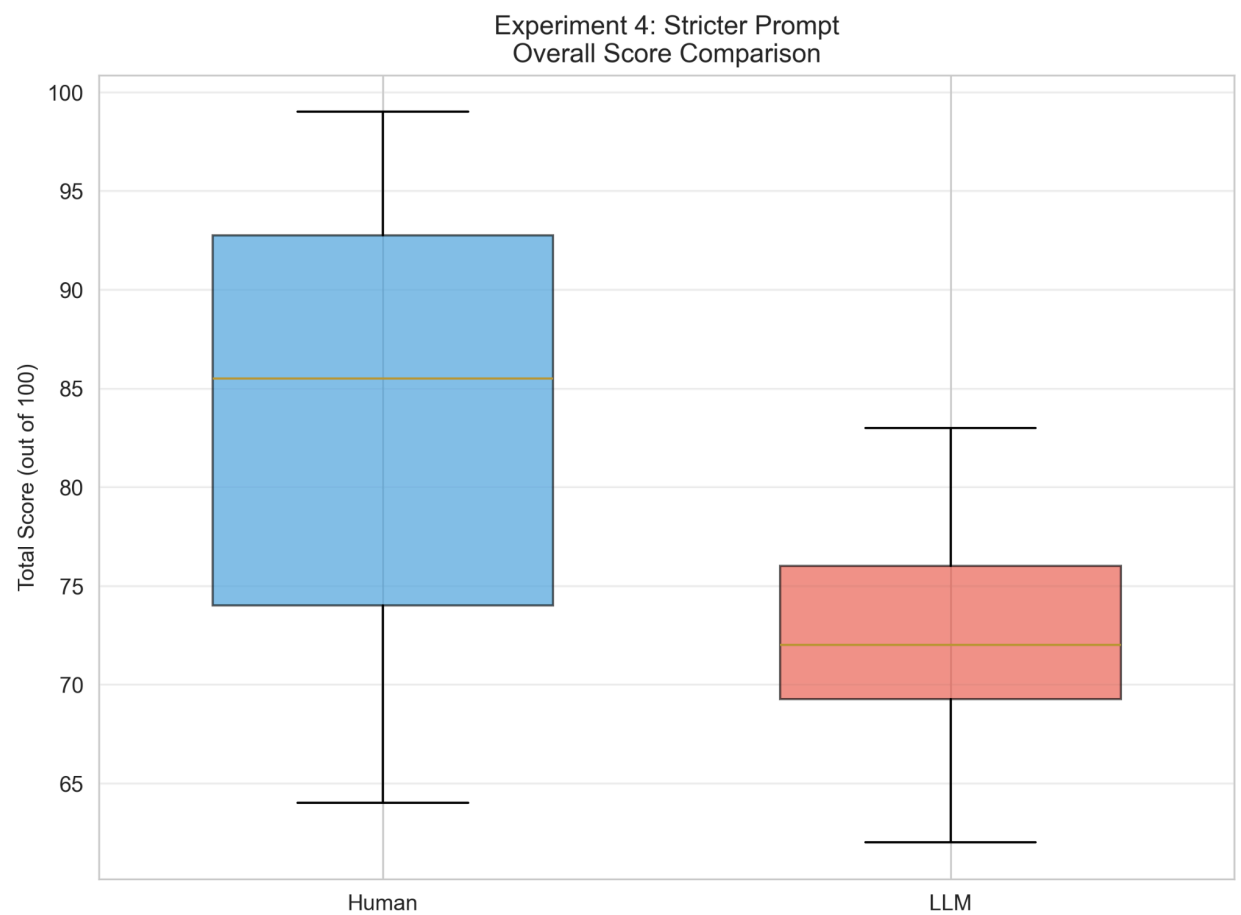
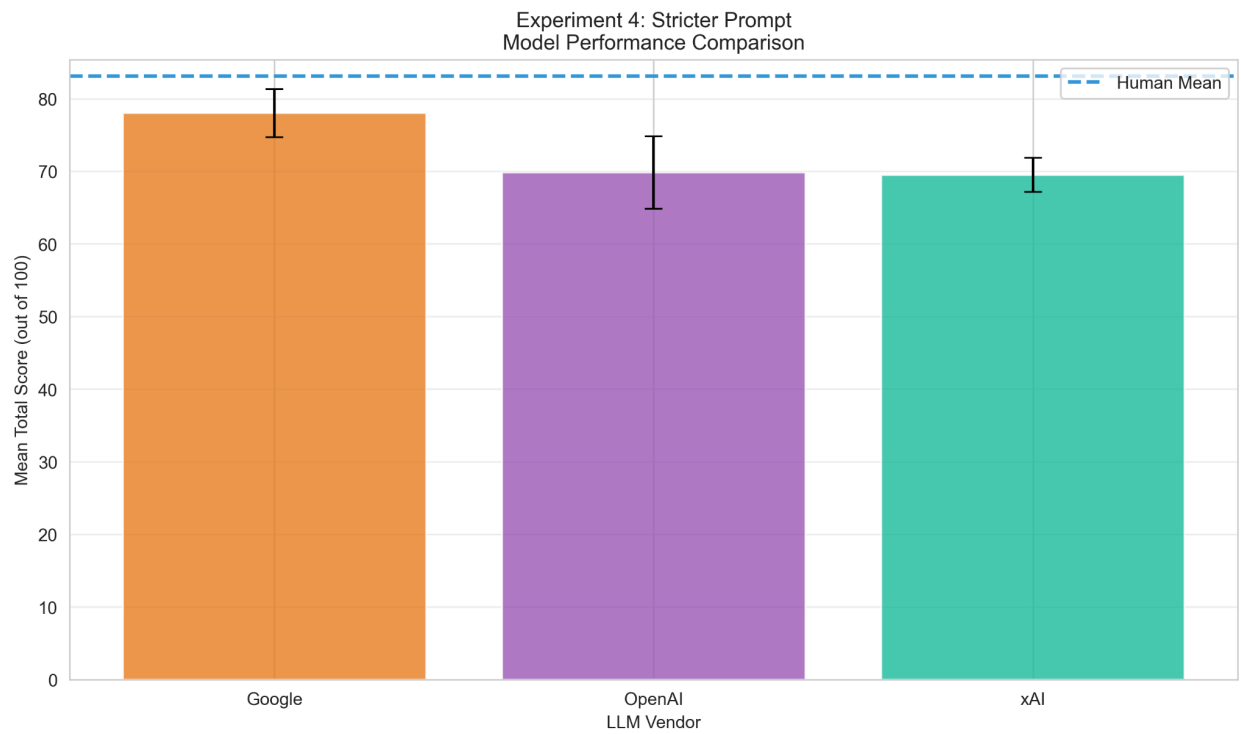
Vendor LLM Performance: All three vendors responded to strict instructions with substantial under-scoring with varying degrees of sensitivity. OpenAI GPT-4 showed extreme response (mean = 69.83, -13.29 from human mean), under-scoring by approximately 16% and representing the poorest vendor alignment observed in any experiment. xAI Grok demonstrated similar conservatism (mean = 69.50, -13.62 from human mean), also under-scoring by over 13 points. Google Gemini showed the least sensitivity to the strict instructions (mean = 78.00, -5.12 from human mean), though it still under-scored significantly. This pattern suggests that while all vendors are sensitive to instructional tone, OpenAI GPT-4 and xAI Grok may be particularly more responsive to directive language (perhaps reflecting differences in model training or instruction-following behavior?). Figure X illustrates vendor performance under the strict Study 4 calibration conditions.

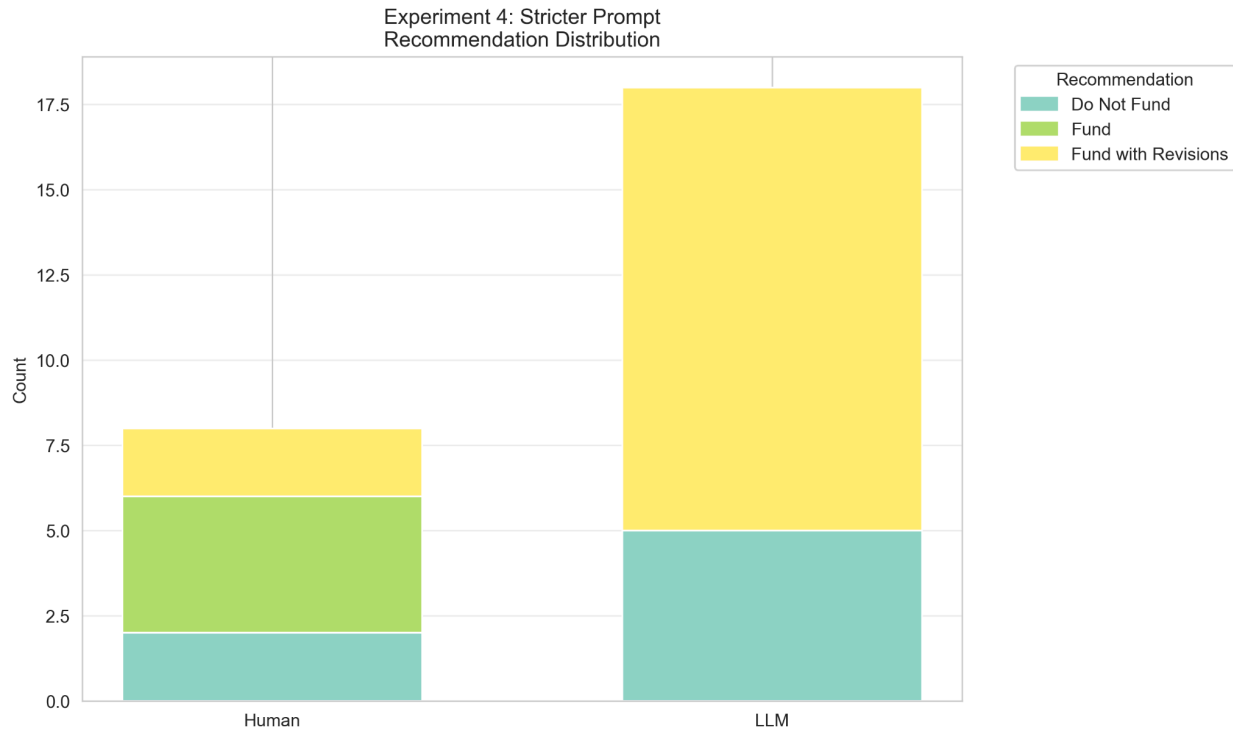
Recommendation Agreement: The Human reviewer distribution recommendations are the exact same (replicated) as Experiment 3: Fund (50.0%), Do Not Fund (25.0%), and Fund with Revisions (25.0%). While the LLMs in Experiment 4 showed increased willingness to issue "Do

Not Fund" recommendations (27.8%) compared to zero in Experiment 1, and maintained a strong preference for "Fund with Revisions" (72.2%), approximately three times the human rate. "Fund" recommendations decreased to 0.0%. Just as in Experiment 1 and 3, we also evaluated agreement at broad agreement and exact agreement. The broad agreement rate was 50.0%, and the exact agreement rate remained 0.0%.

Inter-Rate Consistency: The average standard deviation across the two evaluated applications was 5.19 points for LLMs compared to 12.56 points for humans, representing a 59% reduction in score dispersion. This is nearly identical to Experiment 3, despite the dramatically different mean scores. The strict instructions in Experiment 4 shifted all LLM scores downward uniformly, suggesting that directive language affects score magnitude systematically across all applications, rather than differentially within applications or reviewers.







Discussion

To our knowledge, this study represents the first systematic comparison of prompt engineering strategies for LLM-assisted grant review. Across our three experimental conditions and three commercial vendors, we found several key results. First, prompt engineering profoundly influenced LLM performance. Score differences between experiments ranged from +4.55 points (zero-shot baseline) to -10.68 points (few-shot with strict instructions), translating into a 15.2-point swing based solely on prompt design. Second, including multiple training examples (Experiment 3) achieved near-perfect alignment with human reviewers (-2.57 points, $p = 0.493$), while zero-shot and few-shot combined with explicit strict instructions both failed. Third, related to vendors, xAI's Grok showed the best alignment (+1.46 points overall), while Google Gemini exhibited consistent optimism (+4.16 points) and OpenAI GPT leaned conservative (-3.18 points). Fourth, score alignment did not guarantee recommendation alignment. LLMs showed only fair agreement on funding recommendations ($\kappa = 0.28$), with 59% preferring "Fund with Revisions" versus 25% for humans. Finally, LLMs demonstrated superior consistency (ICC = 0.78 vs. 0.42 for humans).

(Talking point: Seed Grant Term Ambiguity): A potential source of variation in our LLM performance may be due to a lack of an explicit definition of "Seed grant" within our prompts. While the human reviewers at our university likely share an implicit understanding of a seed grant's purpose and expectations from a faculty perspective, the LLMs must infer this construct from its training data, which may represent highly heterogeneous conceptualizations and examples. Additionally, because "Seed grant" lacks a standardized definition, the word "seed" can describe fundamentally different funding philosophies ranging from preliminary data

generation, early-career investigator development, pilot testing of novel methodologies, or high-risk/high-reward exploration. Each of these conceptualizations imply differing levels of evaluation criteria, risk tolerance, and success metrics. Moreover, LLM training corpora likely over-represent publicly available NIH R01 applications, NSF proposals, and large foundation grants, while under-representing internal seed programs that are often smaller and institution-specific. This likely creates a mismatch where LLMs may only evaluate seed grant applications against standards derived from larger funding mechanisms (e.g., that demand higher methodological rigor, more extensive preliminary data, and lower-risk approaches) and not how we have internally defined and operationalized seed grant. Our finding that LLMs systematically over-scored "External Funding Potential" and "Methodological Approach" may reflect confusion about these expectations for seed-level work, specifically related to the criteria "Innovation" and "Methodology", which can be interpreted very differently when evaluating exploratory pilot projects versus fully-developed research programs. Future implementations should include explicit construct definitions in prompts, and not just rubrics and examples. Having clear statements about funding mechanism purpose, risk tolerance expectations, methodological standards appropriate for early-stage work, and success criteria may help calibrate LLM output to better match institutional expectations and human reviewers who share this institutional knowledge implicitly.

(Talking point: Counterproductive Effect of Combining Examples with Stricter Instructions):

Compared to Experiments 1 and 3, the under-performance of Experiment 4 (few-shot learning combined with stricter instructions) was unexpected: our apriori expectation was that by providing both training data and adding instructions to "be critical" and "apply high standards" would help improve the LLM outputs and make their results closer to that of the human reviews but in fact it lead to the largest delta (~ -10.68 points, $p \leq 0.008$) across experiments. This result may reflect a form of compliance bias, which has been previously documented in LLM safety research [citation], where when the LLM is given conflicting signals (e.g., in our case the training data versus the prompt instructions) it will prioritize the explicit prompt instructions over other calibration examples like provided training data examples. In our case, this likely resulted in the broad score reductions across all of the three vendor LLM outputs for Experiment 4, creating worse alignment than with Experiment 1's zero-shot baseline. Conflicting instructions are more difficult for LLMs to handle compared to humans, where human experts can better balance instructions based on their internalized/practiced/SME sense of appropriate score distributions from seeing various examples of well written and poorly written grant proposals in their career. So the dramatic shift from near-perfect alignment in Experiment 3 (-2.57 points) to significant under-scoring in Experiment 4 (-10.68 points) demonstrates how the tone of the instructions can override and negate the training examples. For future research, we thus recommend avoiding the combination of few-shot learning with tone-based instructions (e.g., be strict, be generous, be critical...), and exclusively rely on exemplar/template based calibration with diverse training examples, without any additional directive language. If more conservative scoring is desired, more examples of conservative exemplar reviews should be used rather than instructing the LLM to be more conservative within the prompt instead.